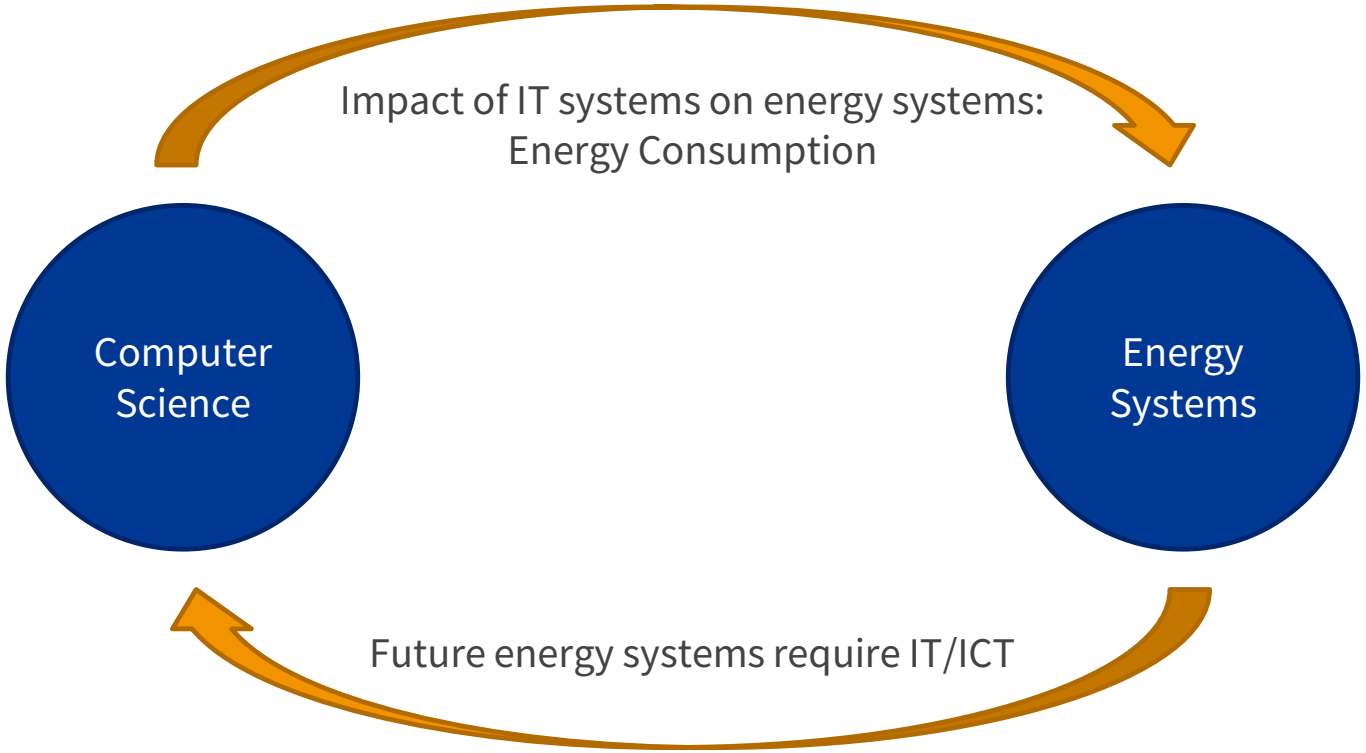
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Energy Informatics I

0. Motivation and Basics
    - Impact of ICT on Energy Systems
    - Energy System Basics
    - Use cases for ICT in Energy Systems
    - Outline of the Course

Computer Networks & Communications
Prof. Hermann de Meer


Impact of IT systems on energy systems:
Energy Consumption
Computer Science
Energy Systems
Future energy systems require IT/ICT



Energy Informatics I

0. Motivation and Basics
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Impact of ICT in Energy Systems

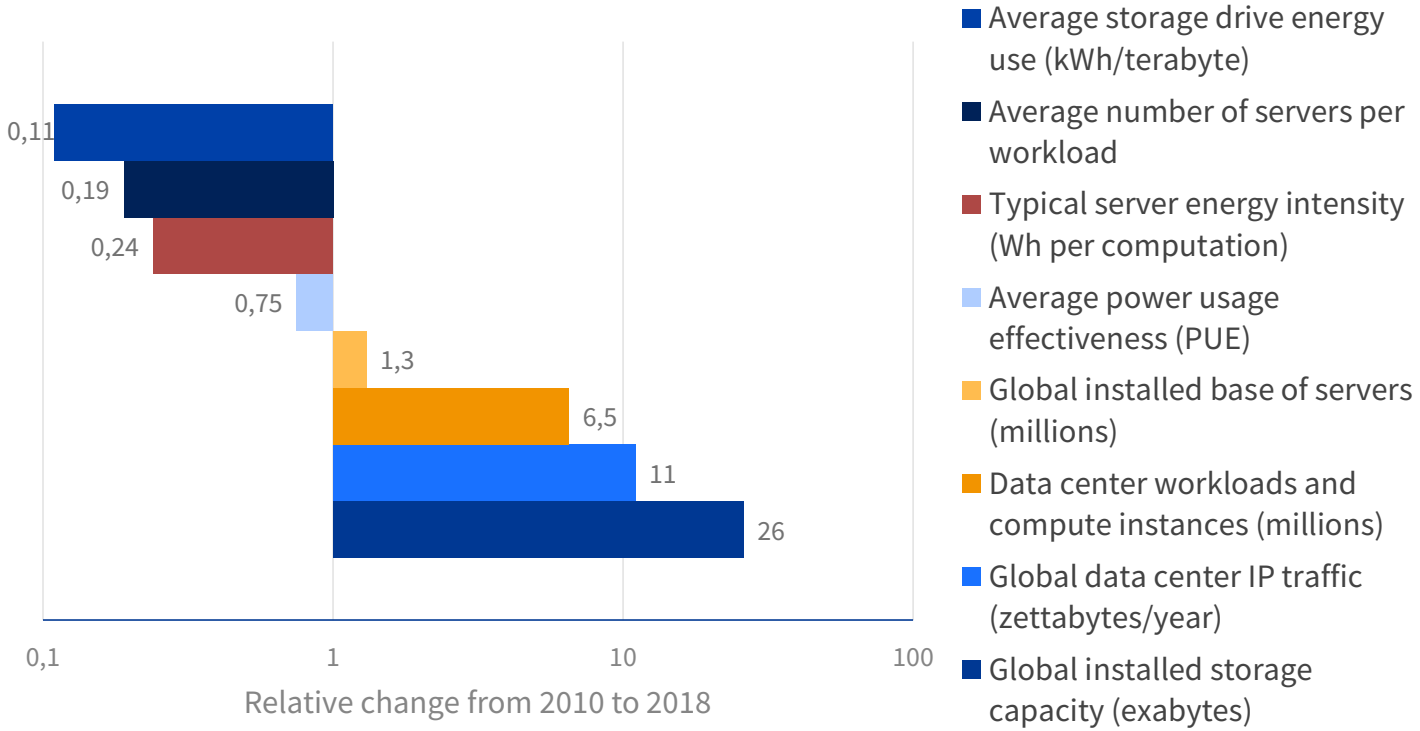
- New technologies require huge amounts of energy
  - **Information and communication technology** accounted for up to **3% of the global energy** demand in 2015
  - Energy efficiency of computers and personal devices have been increasing in recent years due to software and hardware improvements
  - Overall carbon footprint of computer systems has been growing with new applications, require more and more energy:
    - Cloud computing
    - Artificial Intelligence (AI)
    - Blockchain
  - …

Gupta, 2022

Impact of ICT in Energy Systems

Trends in global data center energy-use drivers

(data from: Masanet 2022)



Masanet 2022, Hintermann 2020

Greenpeace #ClickClean – benchmark for energy performance in IT sector (A – best grade):

Video Streaming 2017	Grade
YouTube	A
Amazon Prime	C
HBO	D
Netflix	D
Vimeo	D

Based on data from: http://www.clickclean.org/international

$$PUE = \frac{Total\ Energy}{IT\ Equipment\ Energy}$$

Impact of ICT in Energy Systems

- Energy efficiency of programming languages
  - Mobile devices with limited energy storage require to **focus on energy efficiency of software** rather than on execution time
  - Research results show that a faster language is not always the most energy efficient
  - There is no “best” language considering energy, time and memory usage at the same time.

Based on data from Pereira 2021

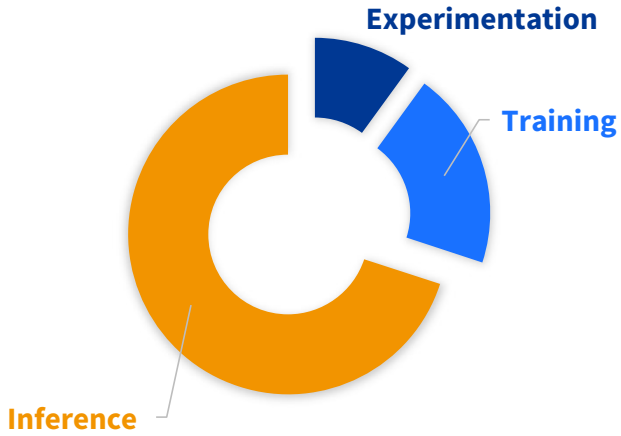
Programming language	Normalized Energy Usage	Normalized Execution Time	Normalized Memory Usage
C	1,00	1,00	1,17
Rust	1,03	1,04	1,54
C++	1,34	1,56	1,34
Java	1,98	1,89	6,01
Pascal	2,14	3,02	1,00
Swift	2,79	**4,20**	2,71
C#	3,14	**3,14**	2,85
Go	3,23	**2,83**	1,05
JavaScript	4,45	6,52	4,59
PHP	29,30	27,64	2,57
Python	75,88	71,90	2,80

Pereira 2021

Impact of ICT in Energy Systems

- Computationally-intensive phases of model development:
  - **Experimentation** – exploration and evaluation of the proposed models, algorithms, training methods.
  - **Training** – evaluation of the models on bigger dataset
  - **Inference** – best model is deployed for production
- Carbon footprint of training large machine learning model is equivalent to almost 400 000 km driven by an average passenger car
- In 2019-2021 Meta's recommendation model sizes have increased by 20 times
- Optimization of algorithms, software, hardware, and infrastructure can reduce the operational carbon footprint by 810 times Wu 2022

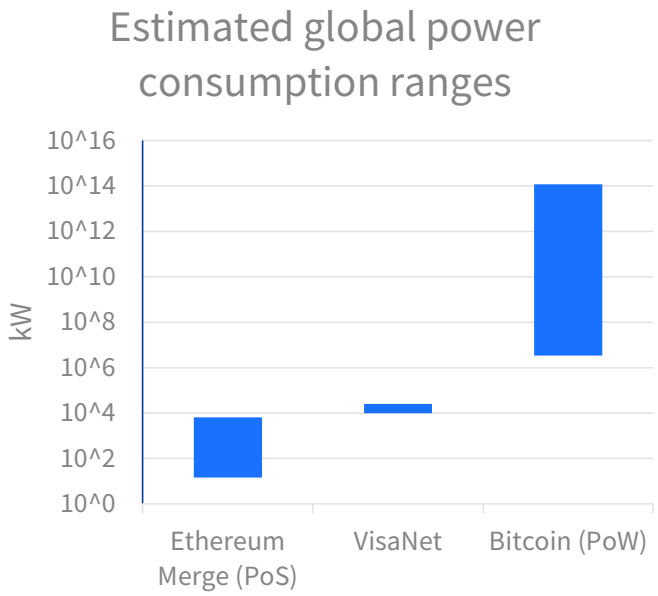
Energy usage of model development phases at Meta



Based on data from Wu 2022

Impact of ICT in Energy Systems

- Energy consumption for producing cryptocurrency depends on the consensus mechanisms
  - **Proof-of-Work** (PoW)
    - Extremely energy-hungry by design
    - 10^{18} hash calculations to 1 transaction in March 2018
  - **Proof-of-Stake** (PoS)
    - A set of validators vote on the next block
    - Reduces energy consumption at least by three orders of magnitude
- Ethereum deprecate proof-of-work on 15.09.2022 and reduced energy consumption by 99,95% (ethereum.org)



Based on data from Platt 2021

Platt 2021, De Vries 2018

- Examples of finalised research projects for energy optimization
  - **SoftAWERE project** by German Environment Agency – develop tools for energy-efficient software and hardware usage
  - **DeepMind's** AI-powered recommendation system reduces amount of energy used for data centre cooling by 40%
  - **RENergetic EU project** – integrate waste heat from supercomputing centre for district heating system
  - **ESN4NW –** Co-location of high-performance computing in wind parks. Optimize high-performance computer operation to maximise wind energy consumption.



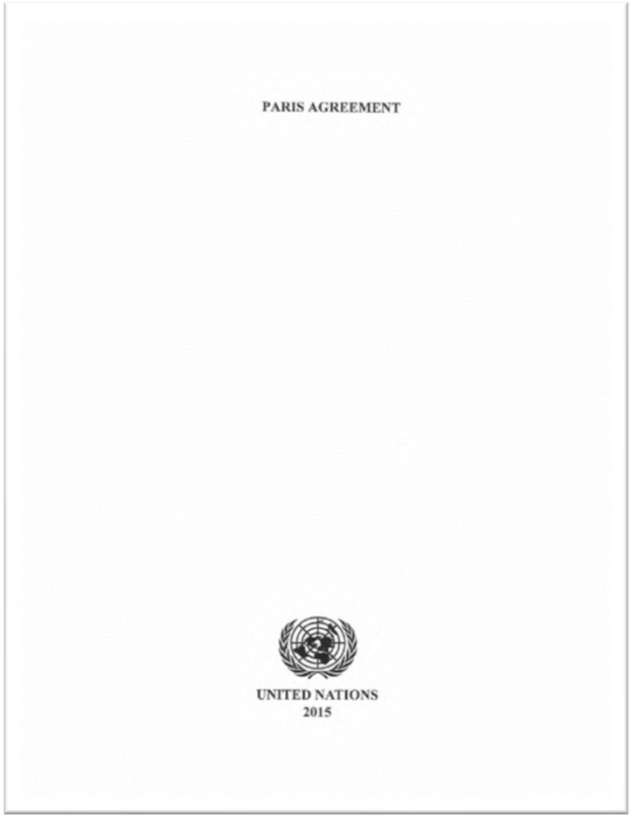
Energy Informatics I

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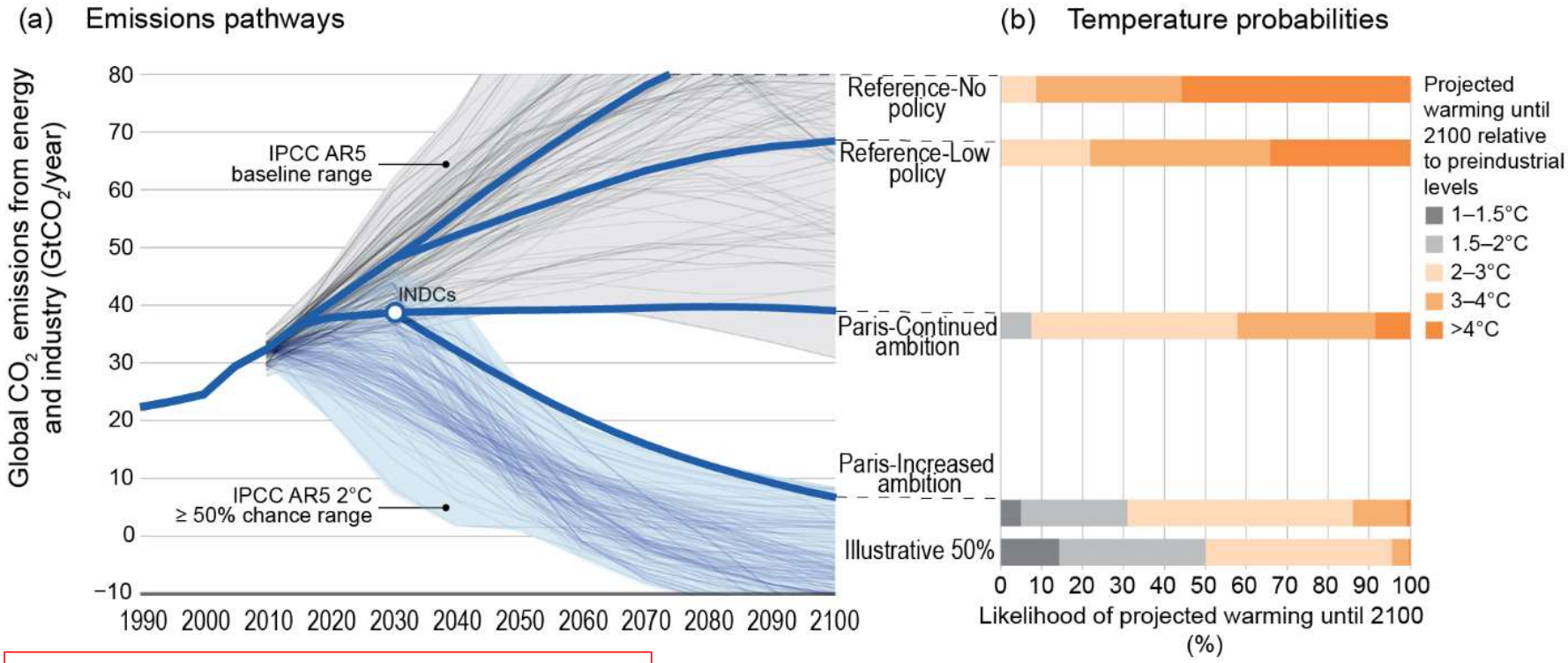
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- Paris agreement goals (2015):
  - Holding global **average** temperature increase **well below 2 °C** above pre-industrial levels and **pursue efforts to limit the increase to 1.5 °C**
  - These targets were chosen to avoid potentially irreversible **tipping points** in the Earth's systems

Avoidable	Probably not avoidable
- Amazon rainforest, Boral forest - Thermohaline circulation (warms Europe) - Sahel - El Niño - Southern Oscillation - East Antarctic ice sheet, permafrost, arctic winter sea ice	- West Antarctic Ice Sheet - Greenland, alpine glaciers - Arctic summer sea ice - coral reefs



Future scenarios and the probability to meet climate goals



INDC: Intended Nationally Determined Contributions
IPCC: Intergovernmental Panel on Climate Change, Fifth Assessment Report AR5, 2014

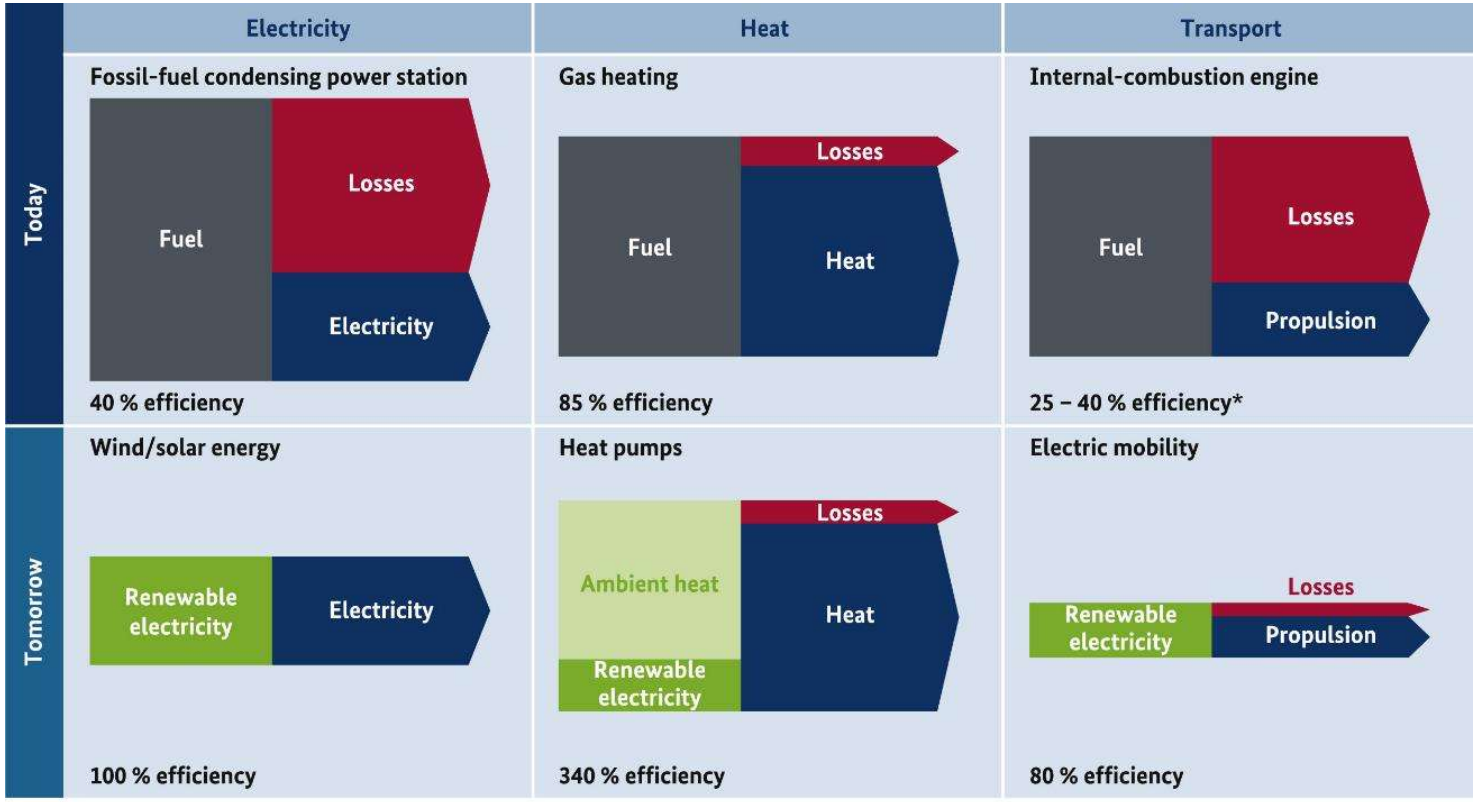
The source for the diagram given in the report is U.S. Global Change Research Program: Climate Science Special Report, Fourth National Climate Assessment (NCA4), Volume I, chapter 14.2. https://science2017.globalchange.gov/ (license: public domain)

- Policy to meet the goals of the Paris Agreement, e.g. European **2020 climate & energy package** (targets for 2020)
  - 20% cut in **greenhouse gas** emissions (from 1990 levels)
  - 20% of EU energy from **renewables**
  - 20% improvement in **energy efficiency**
- The goal was achieved, but with some minor setbacks
  - Greenhous gas emissions **are 31% lower than in 1990**
    - But only 21 Member states reached their national target
  - The EU achieved a **21.3% share of renewables**
  - **25% of reduction in energy consumption** was also achieved
    - **BUT:** The lockdowns due to COVID-19 have a huge impact on that data

https://ec.europa.eu/clima/policies/strategies/2020_en
https://www.eea.europa.eu/highlights/eu-achieves-20-20-20

Energy System Basics

Efficiency of energy sectors



* The efficiency of internal-combustion engines in other applications (e.g. maritime transport, engine-driven power plants) can exceed 50 %.

Tom Brown et al. CC-BY-4.0 https://doi.org/10.1016/j.rser.2018.04.113

- Transmission Grid
  - **Long** distance and **high** voltage levels
  - **Big** power stations and industry
- Distribution Grid
  - Medium to short distances and stepped down voltage levels
  - Small loads
  - **Additional** Distributed Energy Resources (DER) and emerging loads
    - Wind and solar power plants
    - Combined Heat & Power (CHP) plants
    - Electric Vehicles (EV) and heat pumps

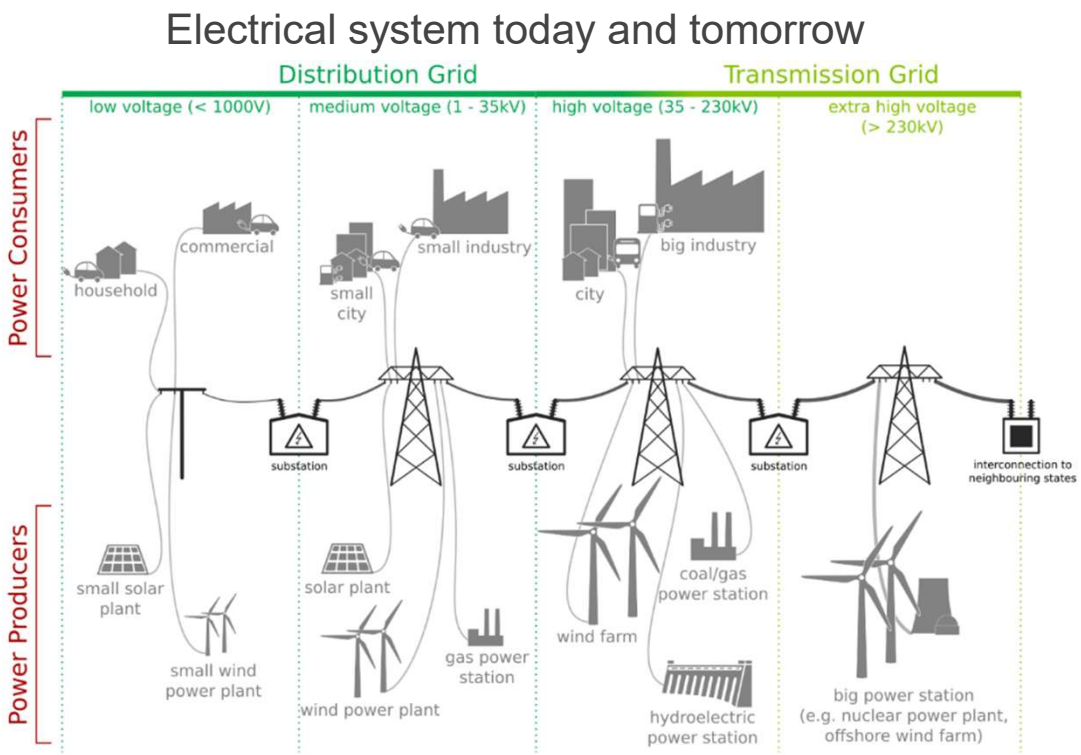
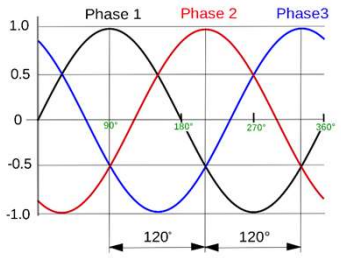
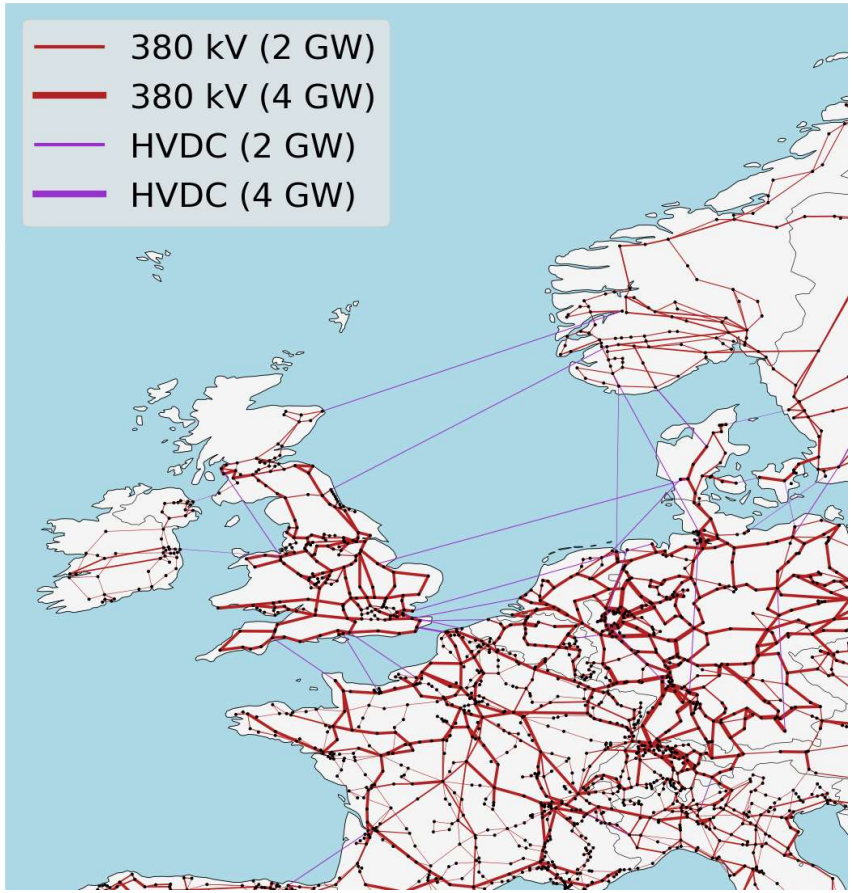


Figure by Philipp Danner, CC-BY-SA 4.0, voltage levels according to IEC 60038

- **Direct Current (DC):**
  - Unidirectional flow of electric charge
  - Typically constant voltage
- **Alternating Current (AC):**
  - Periodically changing direction of current flow
  - Typically **sine wave** and operated as **three phase** systems



Three phase system



Source: https://pypsa-eur.readthedocs.io/en/latest/_images/elec_s.png

- Demand & supply matching
  - European power grid with frequency of 50 Hz (U.S. 60 Hz)
    - Load **>** Generation → **lower** frequency
    - Load **<** Generation → **higher** frequency
  - **Same frequency** everywhere in the grid → Indicator for control
  - Frequency control via **spinning reserve**, **balancing power** (primary, secondary, minute, hour, ...)

    → More details in the chapter Energy Economics

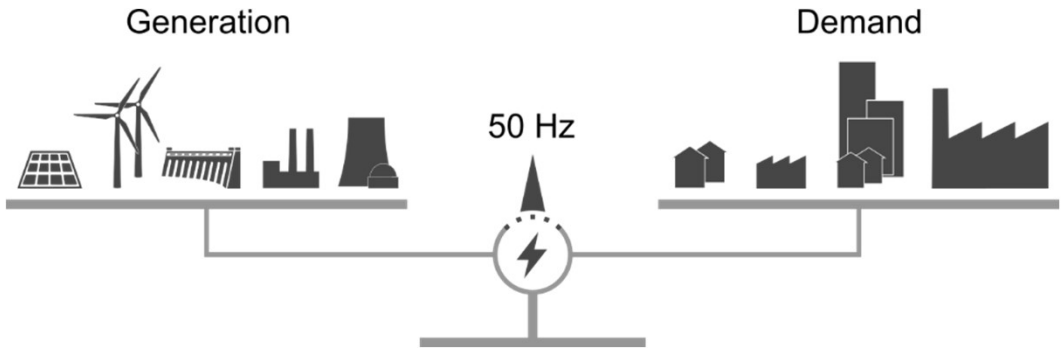


Figure by Philipp Danner, CC-BY-SA 4.0

- Summary
  - Steady **balance** between **demand and supply**
  - Traditional assumptions of energy management and control
    - Demand cannot be controlled
    - Electricity cannot be stored (in the grid)
    - Mainly centralized supply at the high voltage grid
    - Demand located in the low voltage grid
  - Standard control using **spinning reserve**, **balancing power** (primary, secondary, minute, hour, ...)
    - Triggered by deviations of **frequency**

Major Principle: Supply follows demand



Energy Informatics I

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- Implications of renewable energy sources
  - Supply only **partially** controllable or predictable.
  - Supply becomes **decentralized** and moves towards edge side into the low voltage distribution grid
  - Potential **reversal** of power flow

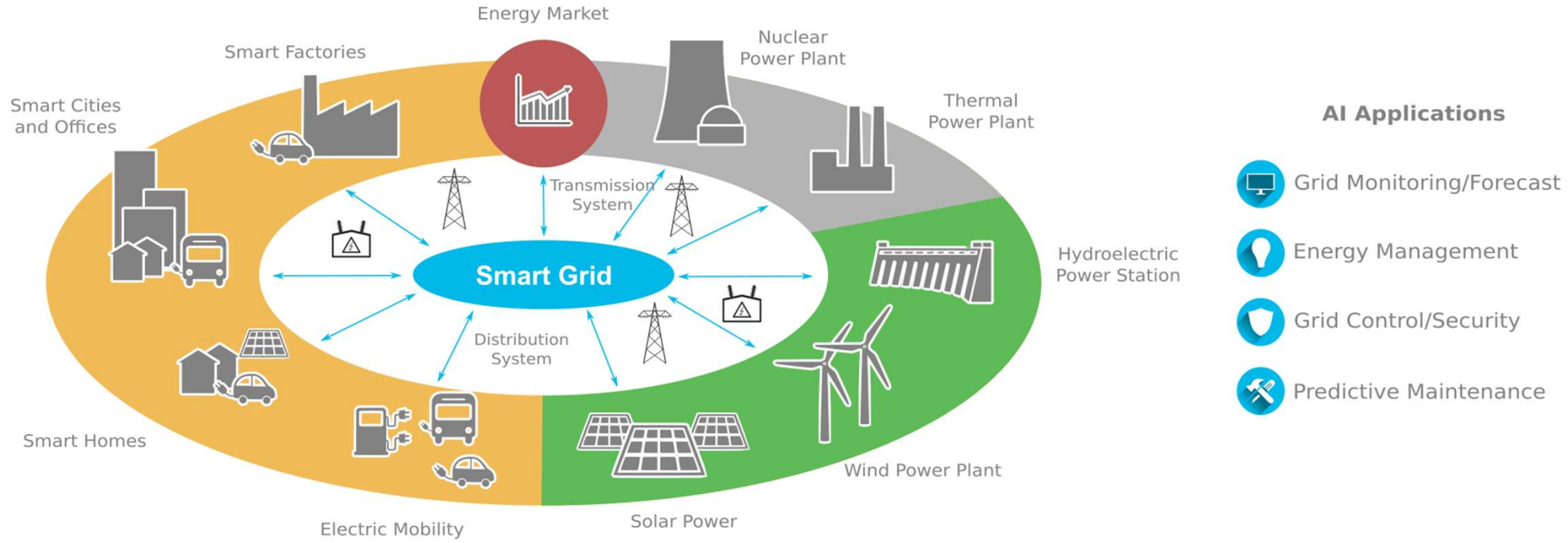
New Challenge: Demand Side Integration!

- Consequence
  - Discover and exploit degrees of freedom for decentralized “demand (and supply) management” (**load shifting**)
  - Develop **new storage** technologies

Need for ICT

Use Cases for ICT in Energy Systems

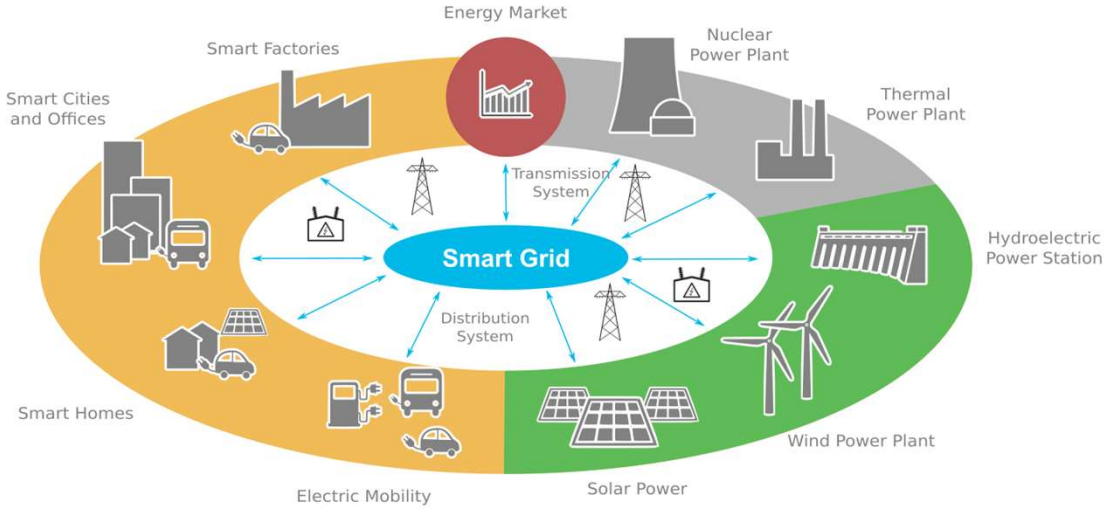
- Shifting from finite generation (grey) to volatile renewable producers (green)
- Smart devices (orange) are used by the smart grid (blue) to operate safely



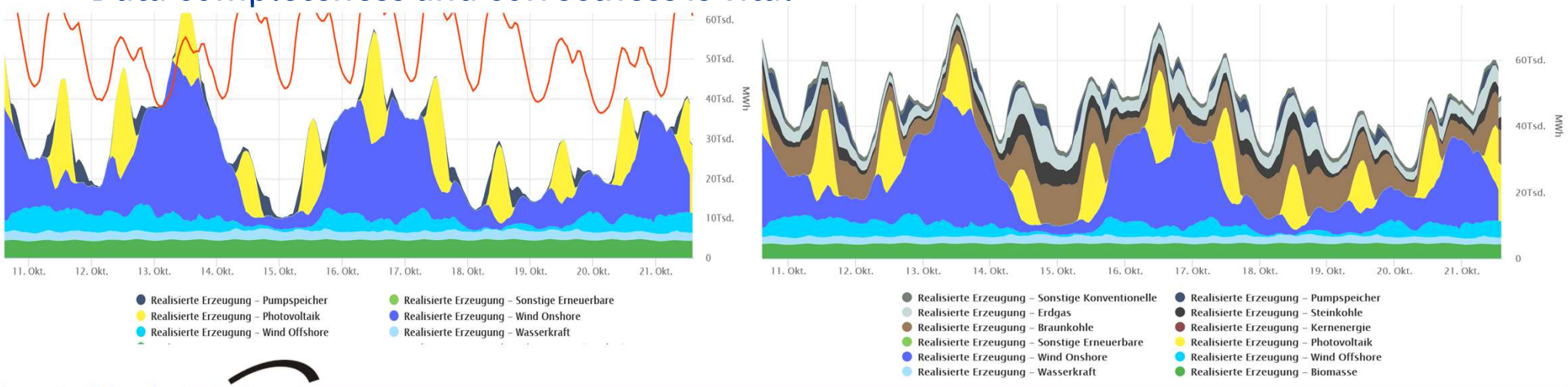
© Philipp and Dominik Danner CC-BY-SA 4.0

Use Cases for ICT in Energy Systems

- A chance for new business actors and business cases
  - Smart Meter Operator (including data communication and storage)
  - E-Mobility Service Provider, Charging Point Operator
  - Smart City
  - Renewable Energy Communities
  - …



- Fluctuation in demand and (renewable) supply limits predictability
- ICT is used to record and store data about generation and loads
- AI methods have been employed in forecasting methods
  - Energy data, e.g., generation profiles, used to train AI models
  - Data may be collected in real time using ICT: Consider Latency!
  - Data completeness and correctness is vital

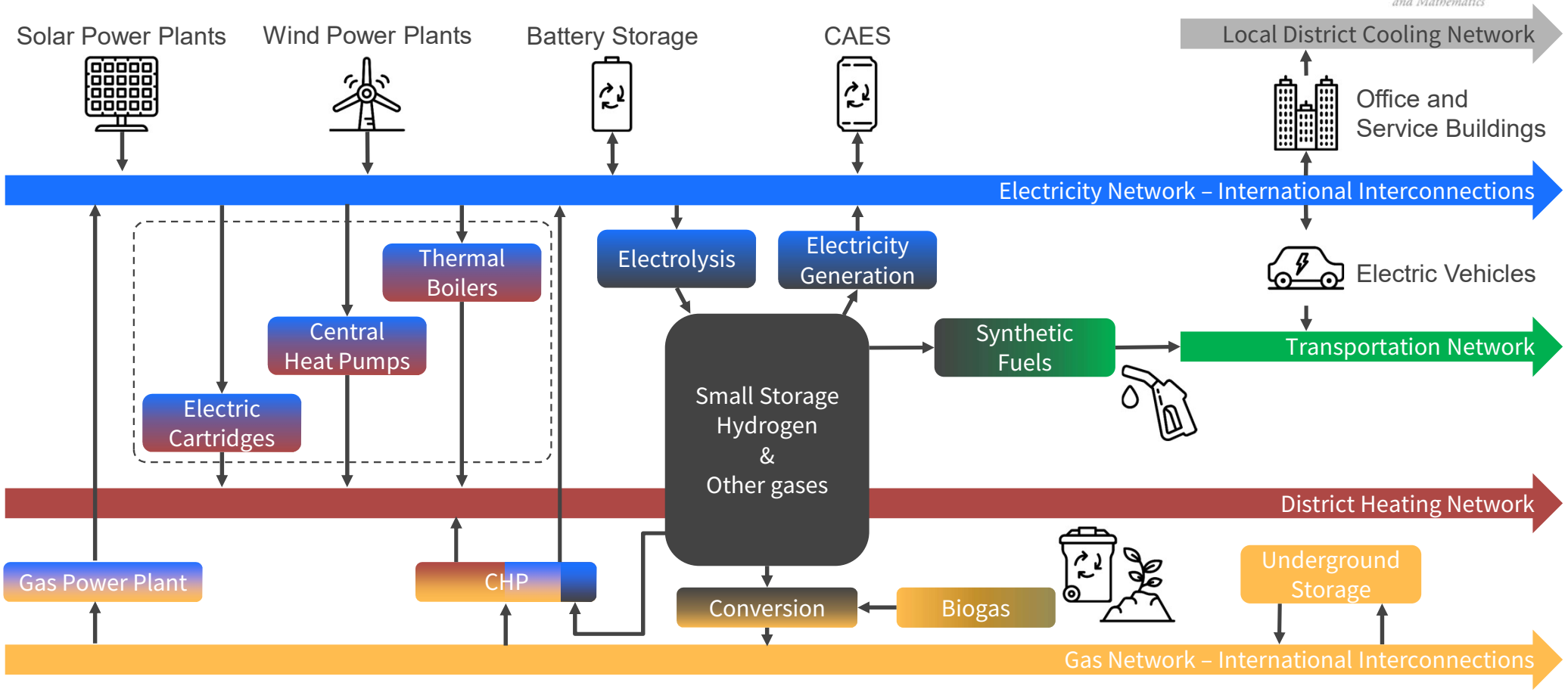


Renewable generation and load in Germany Oct 2024, [https://www.smard.de]

CHP: Combined Heat and Power
CAES: Compressed Air Energy Storage

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Use Cases for ICT in Energy Systems - Sector coupling

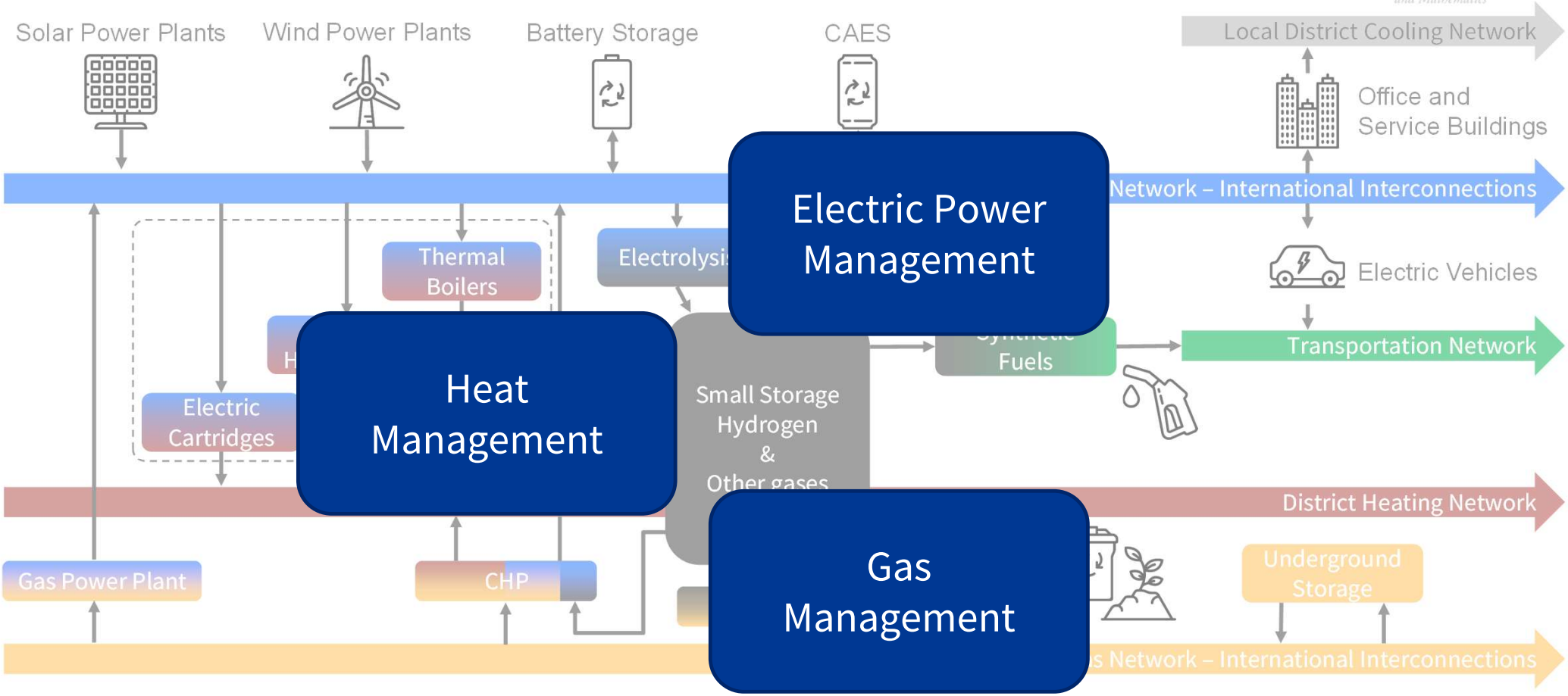


Icons made by Smashicons from www.flaticon.com

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Use Cases for ICT in Energy Systems - Sector coupling

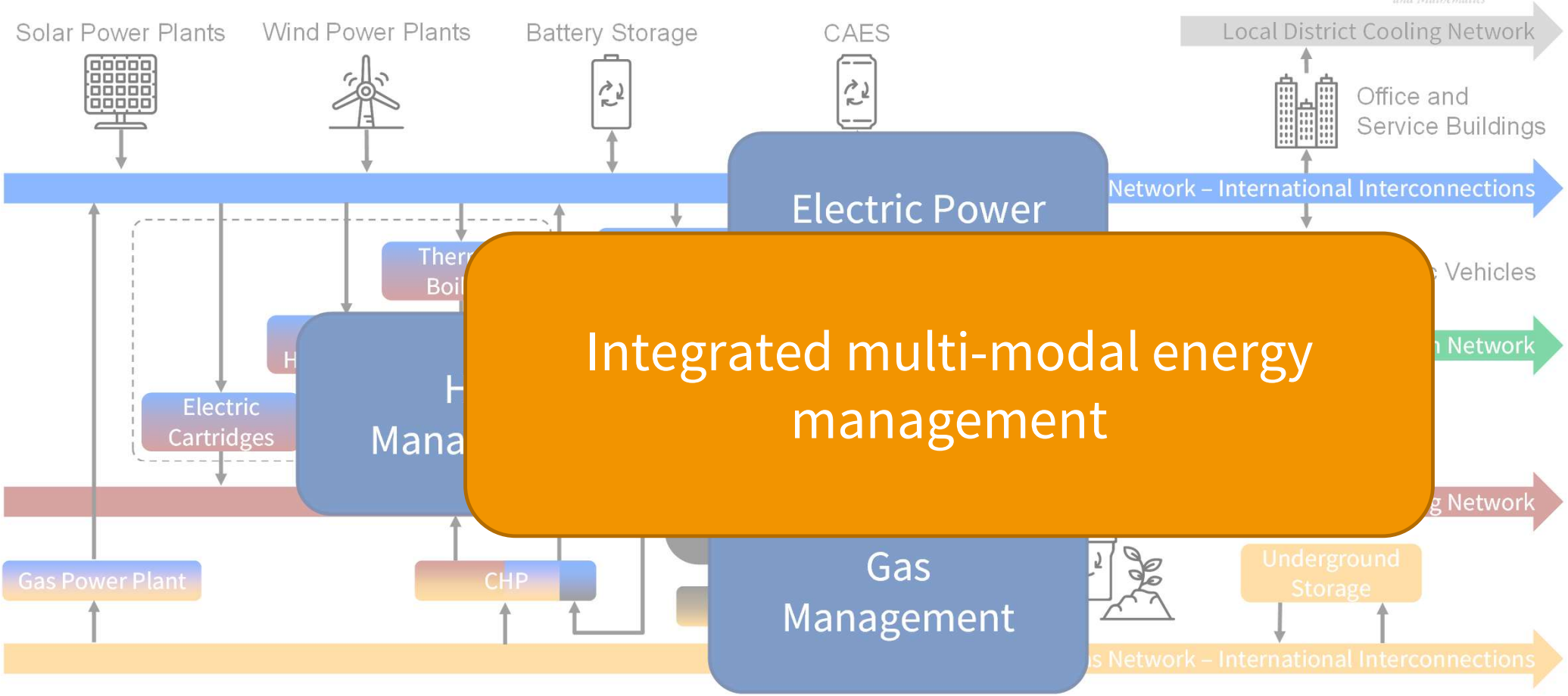


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Use Cases for ICT in Energy Systems - Sector coupling



Icons made by Smashicons from www.flaticon.com

- Interoperability between energy systems (electricity, gas, thermal)
- Integrated energy management in distributed systems to increase **efficiency**, **flexibility** and **stability** of the combined sectors
- Energy conversion between sectors
  - **Real**, e.g. power-to-gas,
  - **Virtual**, e.g. changing conversion rate of electricity and heat in Combined Heat and Power Plants (CHP)
- Additional storage capacity
  - **Real**, e.g. battery, pump storage
  - **Virtual**, e.g. demand shifting

- Interdependency between multiple sectors
  - Failure of the energy system impacts ICT
    - Servers, routers all work with energy
    - Power quality issue even can damage ICT infrastructure (overvoltage)

    → ICT protection mechanisms required (battery systems, overvoltage protection)
  - Failure of the ICT system impacts power systems
    - Legacy grid: rather unlikely, due to low dependency of power system on ICT
    - Future smart grid: Many grid participants will connect to the smart grid, hence high dependency on reliable ICT systems

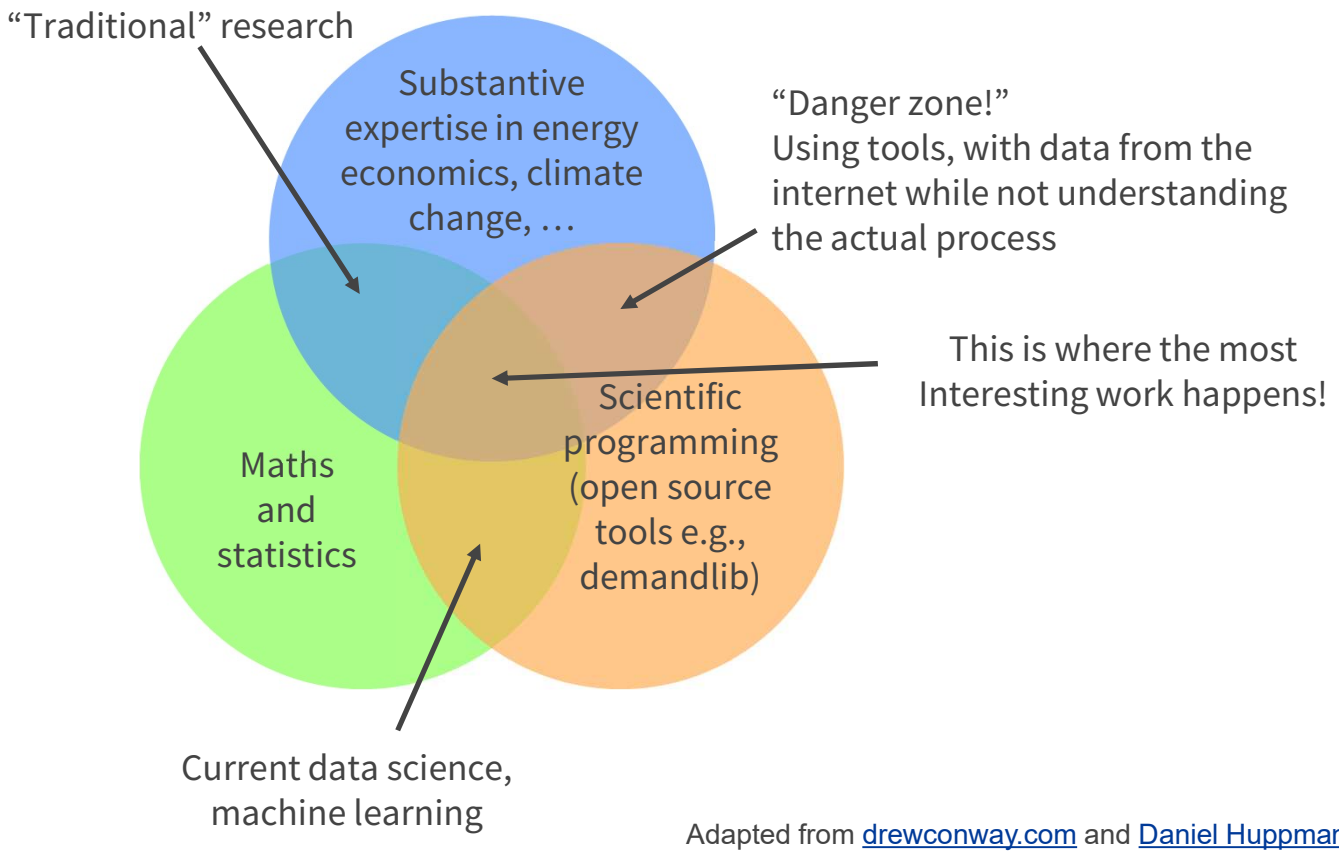
    → ICT-based controlling of the power system needs to be fail safe
  - ICT can even help to recover after brownout or blackout scenario

Use Cases for ICT in Energy Systems

Distributed Energy Resources
Energy Economics
Advanced Metering Infrastructure
Electric Vehicle Charging
Smart Grid Architecture Model
Substation Automation
Smart Converter

DRES
Flexibility

A focus in Energy Informatics II



Adapted from drewconway.com and Daniel Huppmann



Energy Informatics I

0. Motivation and Basics
    - Impact of ICT on Energy Systems
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    - **Outline of the Course**

Outline of the Course

0. Motivation and Basics

1. Fundamentals of Energy Systems
   - 1.1 Distributed Renewable Energy Sources
   - 1.2 Flexibility
   - 1.3 Energy Economics
2. Smart Grid Architecture Model (SGAM)
   - 2.1 Introduction of Smart Grid Architecture Model
   - 2.2 Application of SGAM
3. Communication in Smart Grid
   - 3.1 Standards and Technology
   - 3.2 Advanced Metering Infrastructure
   - 3.3 EV Charging
   - 3.4 Substation Automation
   - 3.5 Smart Converter
4. Conclusion

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